

→ k-Nearest Algorithm (KNA)

(6%) Numeric

45.2% PPM

↳ percentage of microbacterial factor

→ Supervised Machine Learning

Regression

(Numeric Value)

↳ continuous

Class

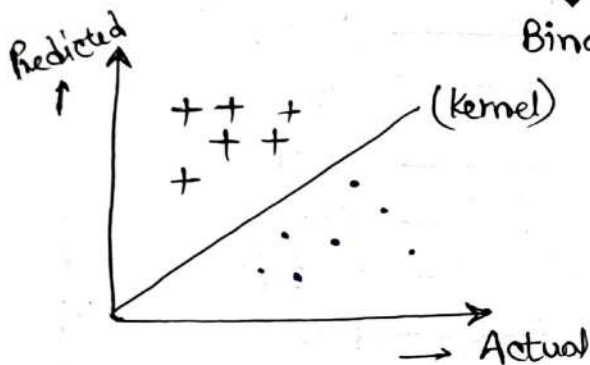
(Categorical Value)

(0/1)

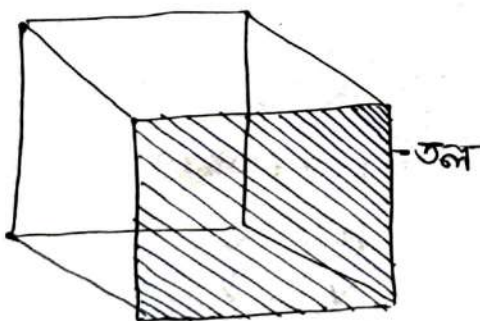
Binary

(2+)

Multi-class



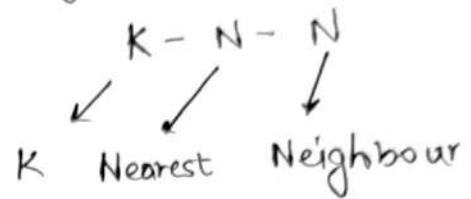
2D Data separation with Kernel



3D Data Separation

→ KNN

→ Weighted KNN



→ How Regression Models are ranked?

↳ via Error Rate

RMSE

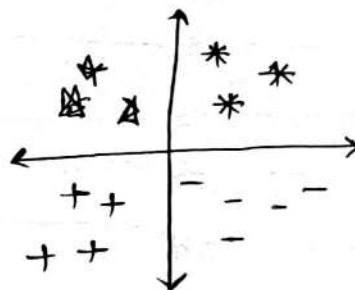
MSE

SSE

R^2

↳ error rate ↓ performance ↑

→ SVM Kernel Separates data



⇒ PCA → Principal Component Analysis

→ Classification Algorithmର କାସ ନାହିଁ

Shashan | ②

→ KNN Algorithm

- ☒ Implactical
- ☒ easy to implement

- ① Select the num 'K' neighbours
- ② Calc Euclidean distance from data points
- ③ Take nearest K
- ④ Among K neighbours, count data points in each category
- ⑤ Assign new data points to the category for which the num of neighbours is max.

Height (cm)	Weight (kg)	Class	Distance D_i	Rank
167	51	U	6.7..	5
182	62	N	13	8
176	69	N	13.41	9
173	64	N	7.615	6
172	65	N	8.24	7
174	56	U	4.12	4
169	58	N	1.414	1
173	57	N	3	3
170	55	N	2	2
170	57	?		

Sample = 10

$$\text{Distance} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

target is fixed
(the unknown)

→

(167, 51)
(170, 57)

Efficient
K-selection

$$\frac{\sqrt{N}}{2} = \frac{4K}{2} = 2K$$

Let $K=5$
Top 5 Ranking

N, U
↓ ↓
3 2
 $N > U$
= ③ → N

When KNN failed? → Very large dataset

① No way to determine best value of K .
Most preferred is 5

② A very low K can be noisy

③ Large values are good but might find difficulties.

④ We use a test set to estimate error rate of the classifier

$$K = \frac{\sqrt{n}}{2} \quad [n = \text{number of examples}]$$

→ Chinese Algorithm

→ Mathematical Dynamics

→ Game Theory

→ VHDL Instruction Programming Language
↳ Rust

→ Advantages of KNN

→ Disadvantages of KNN

→ Weighted KNN

$$\text{Weight}_i = \frac{1}{d_i}$$

$$\text{Weight}_i = \frac{1}{d_i^2} \quad \left] \text{Low Error Rate} \right.$$

→ Lab Topic

Linear Regression

Polynomial Regression